# 29. EMBRYONIC FOLDING

DURATION : 59:59

STUDY SHEET


## COURSE SUMMARY

In this video, we delve into the crucial process of embryonic folding, revealing how the notochord influences the structural development of the embryo. We explore the central role of the notochord in establishing polarity and forming the neural tube, while also discovering the dynamics of differential growth between the ectoderm and mesoderm. This process leads to the creation of the neural groove and pre-aortic vessels, essential for embryonic blood circulation.

Furthermore, we analyse how the flexion of the embryo around its vascular system, using the heart as a fulcrum, is similar to natural and comfortable movements. This complex interaction between different embryonic tissues and structures is key to understanding joint formation and body organisation, offering essential insights for students and practitioners interested in biodynamic embryology and osteopathy.


## 29. EMBRYONIC FOLDING

We will continue with the movement of the embryo and address **folding**. This concept was a revelation for understanding what an **articulation** is. It is the algorithm that generates this **differential growth rate**, leading to the flexion of the embryo.

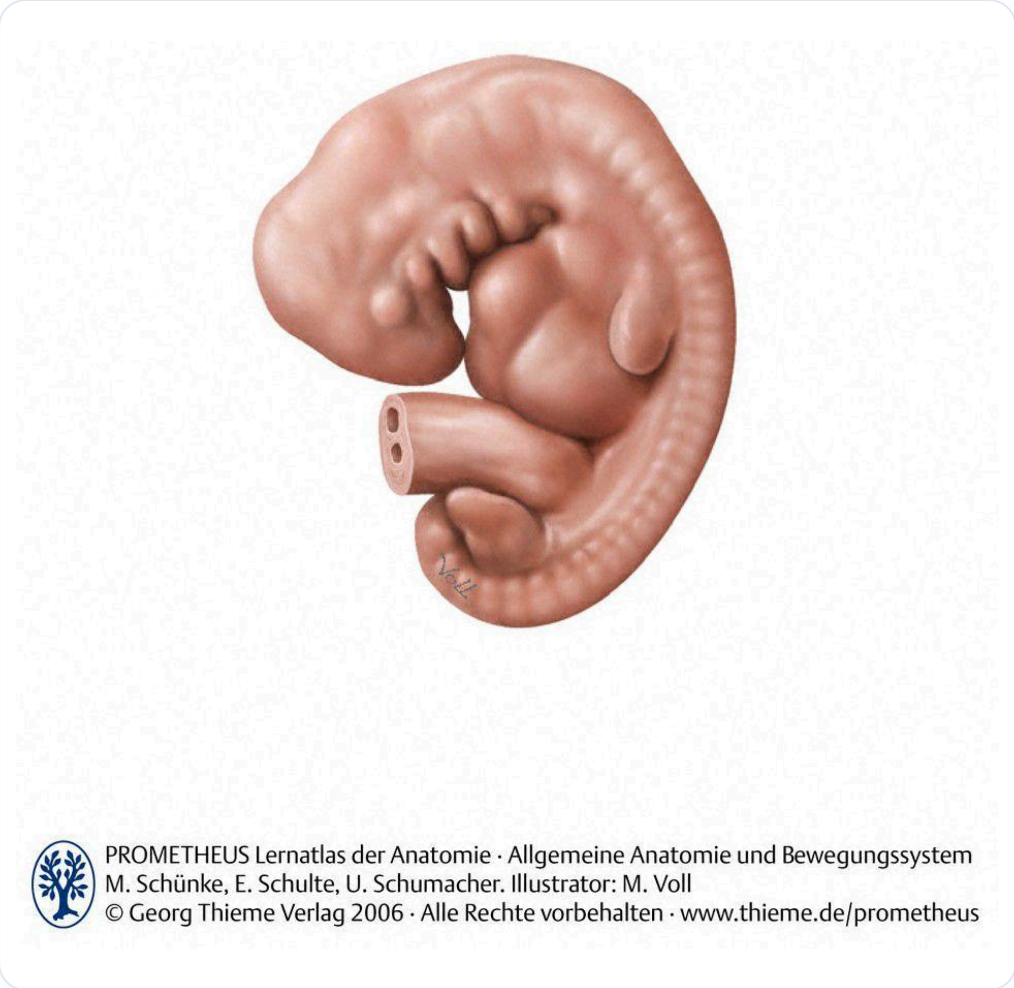


*FIG. — Human embryo, early stage*

A **fluid dynamic** organises the embryo at this level. The **notochord**, essential for its development, will now influence a structure above it: the **neural tube**, or rather the future neural tube.

## THE INFLUENCE OF THE NOTOCHORD

We are between the **eighteenth** and **twentieth day** of development. We have the **axial process**, i.e., the notochord, and above it, the **ectodermal** tissue. We are progressing in the study of embryonic stages.

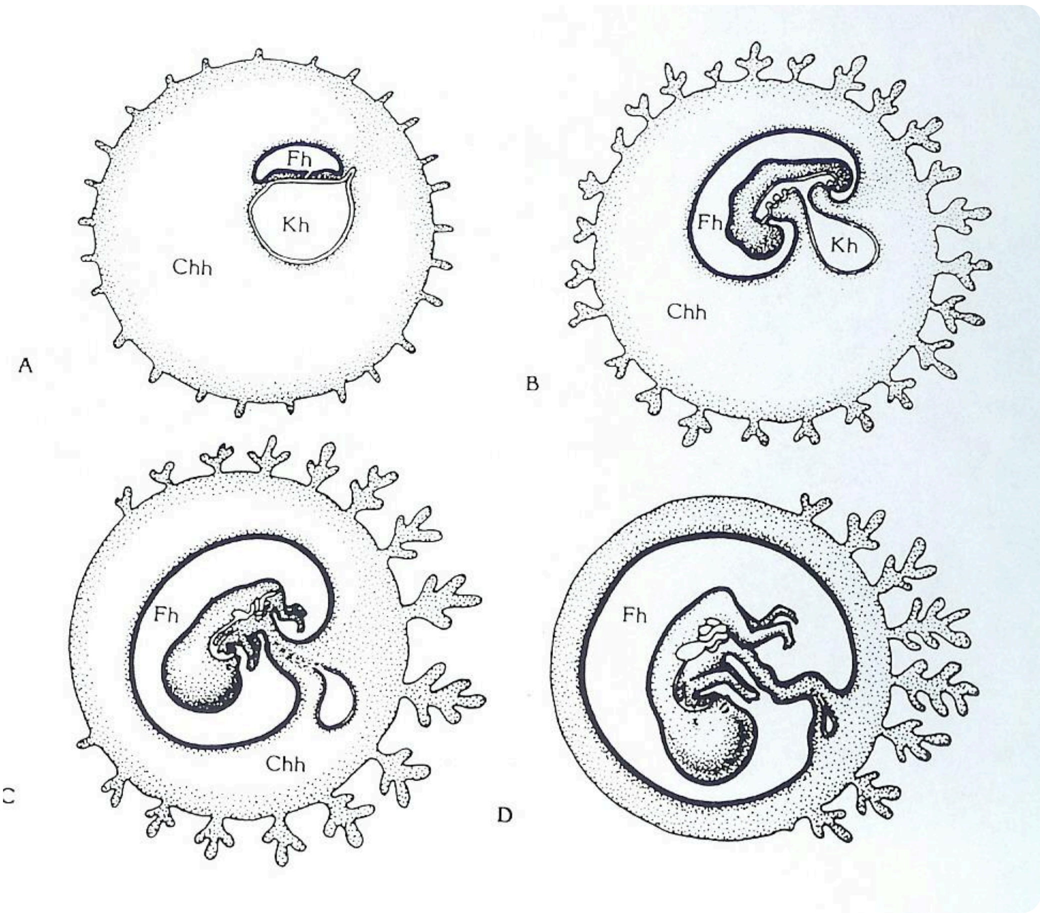


*FIG.* — *Bird embryonic development*

The notochord, which has established **polarity** and created a future space for the **sacrum** and the **base of the skull**, influences the development and structural organisation of the ectoderm. Above the notochord, we observe a **neural plate**.

Let's take a cross-section. The ectodermal cells directly above the notochord receive **inhibitory genetic information**, leading to a slowing of their growth.

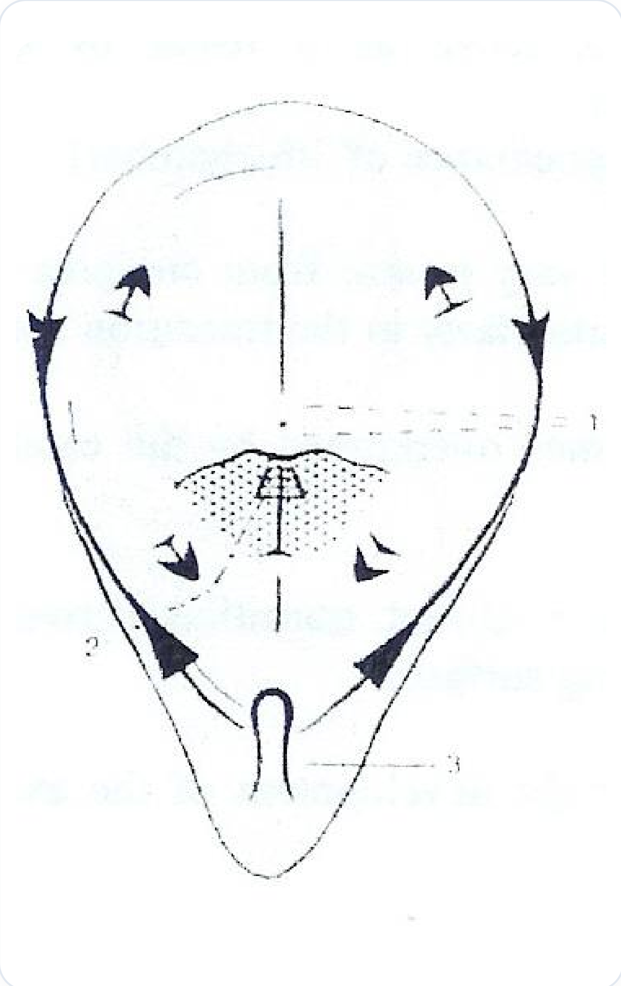

*FIG. — Gastrulation*

- The central cells at the level of the notochord are inhibited in their growth.
- The lateral ectodermal cells, however, do not have this brake and even develop faster.

This process leads to a **change in the shape** of the ectoderm.

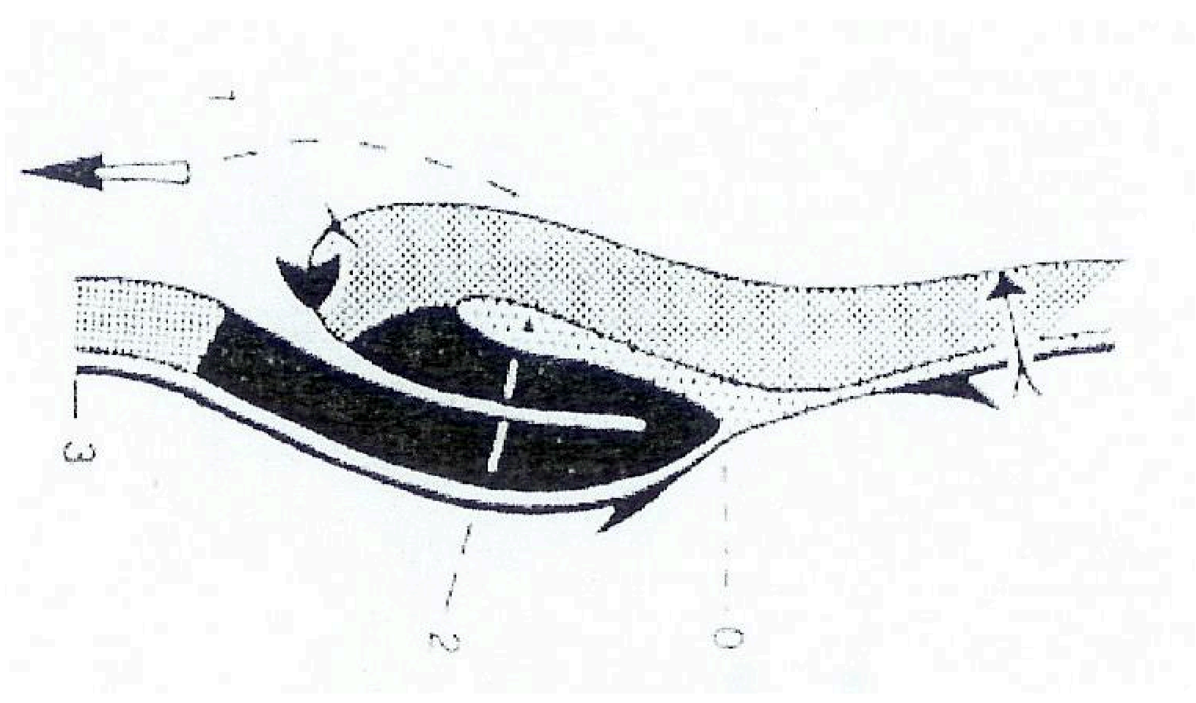

*FIG.* — *Amphibian gastrulation*

## FORMATION OF THE NEURAL GROOVE AND PRE-AORTIC VESSELS

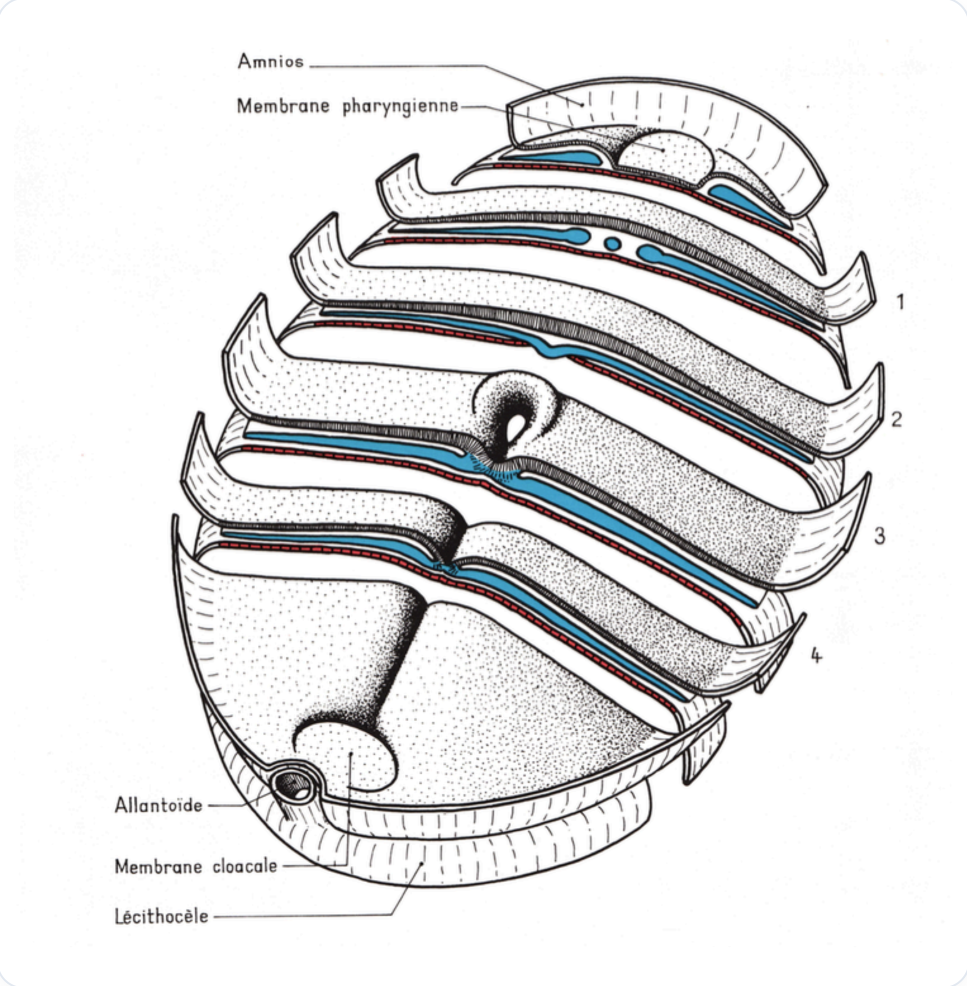


*FIG. — Embryo in sagittal section*

This differential development causes the formation of a **groove**, called the **neural groove**. The lateral cells of the neural plate develop more rapidly, creating this excavation in the ectoderm.

Simultaneously, the surrounding **mesoderm**, little modified and quite liquefied, undergoes reorganisation. We observe the formation of fine **pre-aortic capillaries** in the form of vesicles, which require a specific trajectory and metabolic concentration.

These vessels organise into two mesodermal fluid conduits. The differentiated growth rate between the ectoderm and the mesoderm is crucial. The ectoderm, a large consumer of information, grows very rapidly.

## EMBRYONIC FLEXION AND THE ANGELIC ZONE

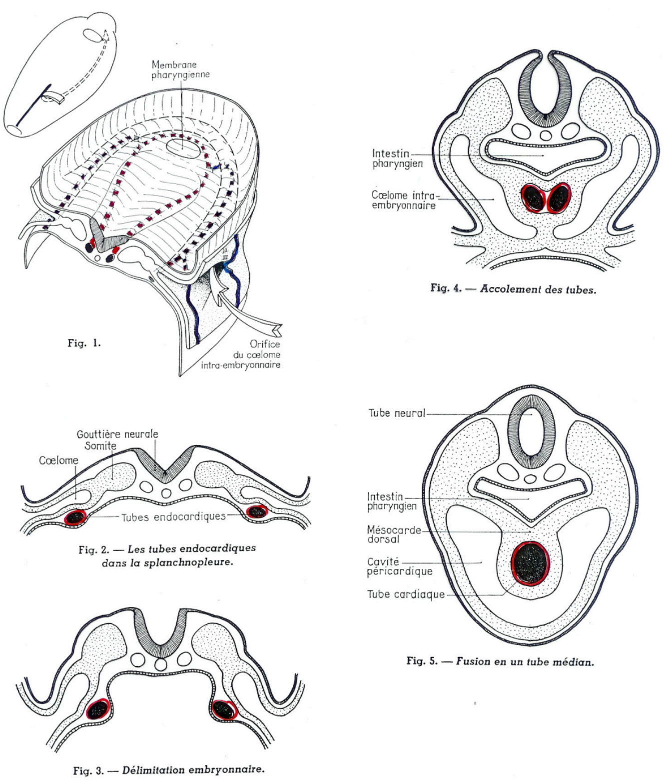


*FIG. — Heart tube development*

At some point, the embryo will perform a **flexion movement** forwards. This movement is fundamental.

The neural plate begins to organise. On either side, the pre-capillary aortic mesodermal vessels appear. They create a **congestive zone** above the neural plate, a small "pond" in front, supplied by two small lateral vessels. This is called the **superior angelic zone**.

This angelic zone will undergo many modifications. It moves towards the midline to form a tube due to the growth of the embryo. The ectoderm (neural plate) is the engine of this growth, while the zone of the aortic capillaries and the pre-cardiac zone act as a **relative brake**.

## THE ROLE OF THE HEART AS A FULCRUM

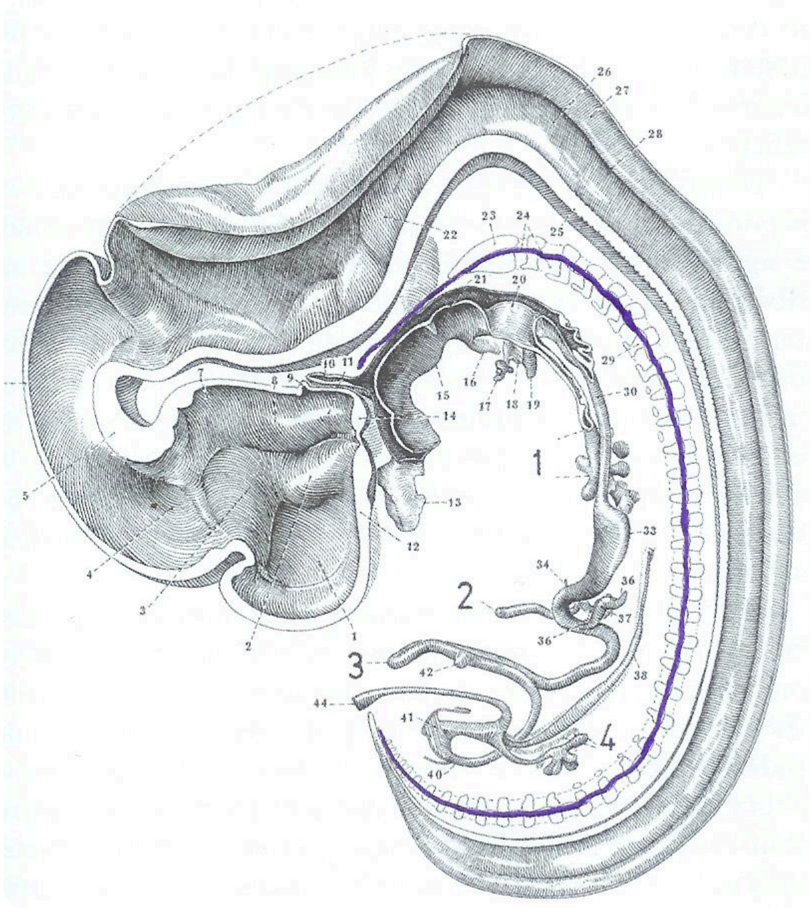

*FIG. — Human embryo sagittal*

The embryo grows, but it encounters a brake imposed by this pre-cardiac zone. What becomes a fulcrum? The **heart**.

The embryo flexes, coiling around its vascular system, which acts as a **pivot**. This movement is often presented as flexion or folding, but it is actually an **expansion** around this vascular anchor point.

Our true initial postural axis is a **vascular axis**. We coil around it. This principle organises all the articulations of the body. If a path is blocked (no developed artery), growth adapts and takes another route.

This movement is similar to that of a baby coiling in on itself, around its vascular system. It is a position of well-being, reproduced in adulthood in the foetal position in case of discomfort.

## ENGINE OF GROWTH AND TROPHIC INFORMATION


*FIG. — Vascularized human embryo*

The ectoderm is the **engine of growth**. It is supplied by the **fluid mesoderm**.

- **Engine of growth:** The ectoderm.
- **Trophic information:** It comes from the endoderm, via the mesoderm, and from the mother via the cord and placenta.

The overall coiling of the embryo at this stage will define specific aspects of development.

## THE CEPHALISATION CASCADE

**Cephalisation**, i.e., the development of the head, leads to a series of processes:

1. **Cephalisation**: The process of cerebralisation of the cerebral vesicle coils and triggers the sequence.

2. **Cardialisation**: Cerebralisation initiates cardialisation, the flexion movement positioning the heart. The heart, initially higher, descends to this location due to growth.

3. **Diaphragmatisation**: The heart presses on the supero-internal part of the vitelline vesicle, creating pressure that forms the **septum transversum**, the primitive anlage of the diaphragm.

4. **Hepatisation**: The septum transversum leads to a phase of sub-diaphragmatic congestion, forming the mesodermal anlage of the hepatic zone.

5. **Adrenalisation and Renalisation**: Later, during straightening, absorption spaces create the adrenalisation and renalisation zones.

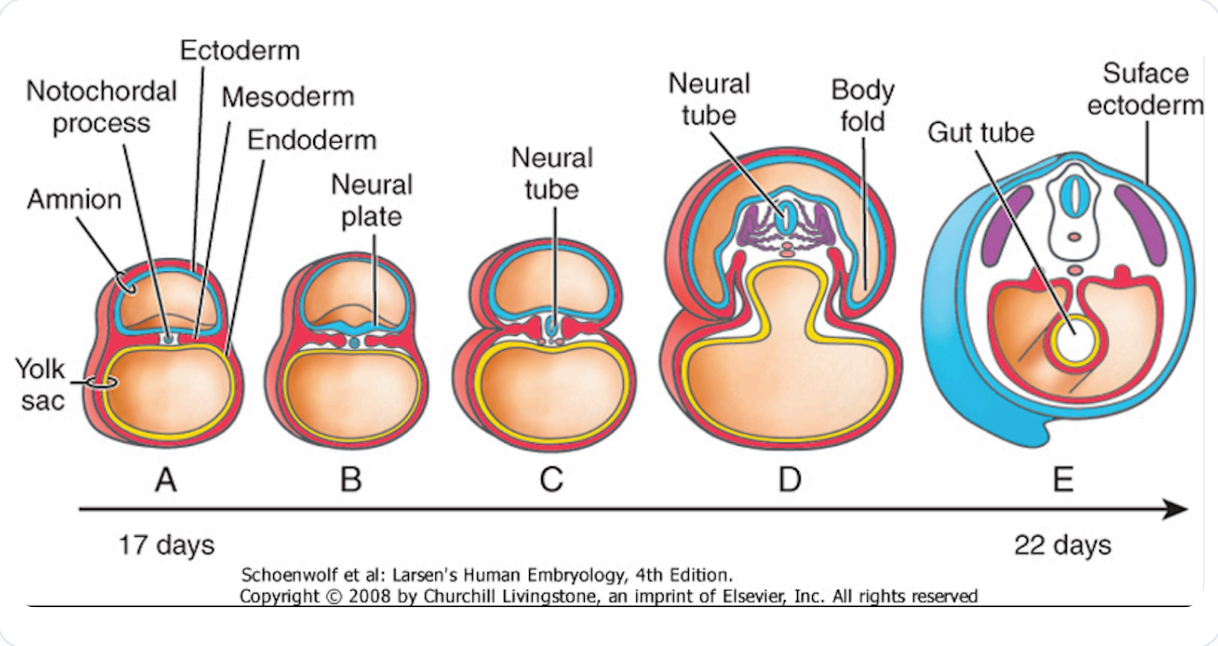


*FIG. — Early embryonic development*

## THE LIVER AND CONGESTION

**Congestion** represents the continuous flow of information from the **embryonic pedicle**. The liver is initially formed only by the waste and exudates of the embryo. A global accumulation of these exudates creates the first impulse of congestion.

The continuous flow of nutritive information from the embryonic pedicle leads to a more significant phase of congestion under the diaphragm, impelling the development of the liver.

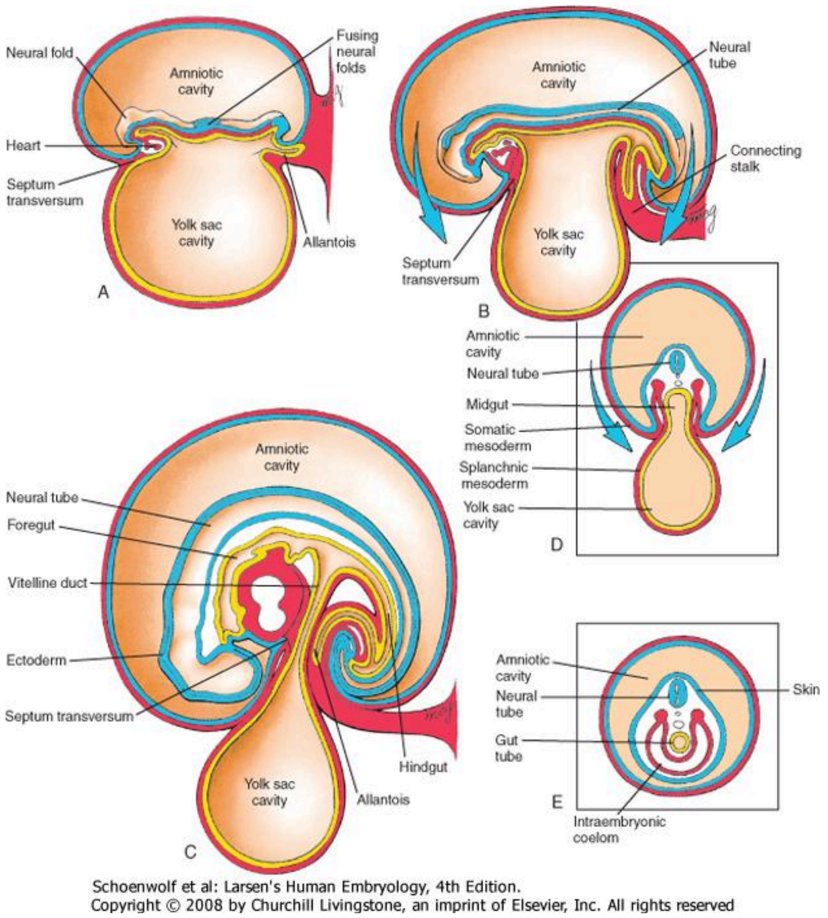


*FIG. — Lateral embryonic folding*

The liver is composed of **mesoderm** (for its vascular plan) and **endoderm** (for its digestive plan). The meeting of these two tissues forms an island. This primitive sub-diaphragmatic congestion is the first vascular impulse.

## VASCULAR DEVELOPMENT

The embryonic pedicle and vitelline systems provide trophic information from the mother via the placenta.

Initially, **primitive aortic vessels** are distributed homogeneously. Then, the embryo segments via the venous system, through a process of assimilation and dissimilation.

Explanatory diagram

*FIG. — Explanatory diagram*

A venous network, composed of the **cardinal**, **subcardinal**, and **supracardinal veins**, forms a vast central vascular network (vena cava, portal veins, etc.).

At the aortic level, the flexion movement of the embryo fuses the two primitive aortas into one.

Two large vascular networks develop almost simultaneously:

- An aortic network.
- A venous network.

It is essential to understand that initially, the embryo coils around these **primitive mesodermal vascular axes**.

## THE CRANIO-SACRAL-STERNAL COMPLEX

This mechanism is fundamental for **occipital** and **sacral** development. The heart is a central point around which everything coils. This story is repeated at the level of the sternum.

This is why we speak of **cranio-sacral-sternal treatment**. It is a movement of flexion, internal rotation, and adduction. It is a developmental movement towards the centre, resulting from differential growth.

The occipital system, the sacral system, and all lines of force reverberate at the level of the sternum. The sternum is a very **somato-emotional** zone.

## EVOLUTION OF THE STERNUM AND DIAPHRAGM

Initially, there are two **sternal bars** that meet at the manubrium. This junction forms the **angle of Louis**, which appears when the mediastinum reaches equilibrium.

Later, the **xiphoid process** forms. It expresses the equilibrium of the peritoneum when the abdominal cavity is completely integrated and the intestine has completed its rotation.

The xiphoid process therefore symbolises the equilibrium of the peritoneum and the entire history of its development.

A point of equilibrium for the mediastinum is often located around **T3-T4**, as these vertebrae share a common mesentery with the aorta and part of the digestive system. For the peritoneum, **T12** is the axis of rotation of the digestive system.

The body is a complex knit. In **biodynamics**, we work on the eight fields. When the occiput descends and the sacrum descends (flexion), the sternum ascends. This **"whiplash"** movement is often associated with a loss of decision-making power.

If the sacrum ascends when it should descend in flexion, this creates a misalignment in the movement.